Applied Statistics Lab for the Social Sciences

Valentina Lorenzoni & Chiara Seghieri

Sant'Anna School of Advanced Studies

Data cleaning and management

Basic rules:

- Preserve original data (work on a copy of the original dataset).
- Keep data cleaning and manipulation code separate from the analysis code.
- Include sufficient explanation of the process and number of observations retained at each steps. Examples of data manipulation include dropping, adding, or editing data.

Data cleaning 1

Data cleaning is an important step before doing any analysis.
Cleaning data can be time-consuming, but it can save a lot of

time and energy down the line.
Principal steps:

- Look at the distribution of every variable (mean, max, min,...).
- Issues might be due to: typos (i.e., age 200 years), outliers (both plausible and implausible), time-invariant variables not being time-invariant, etc.
- Check for duplicates in unique IDs.

Data cleaning 2

- Missing values: check the share of missing observations for a given variable. Also pay attention on how they are coded (i.e. -9, 999, -8, ...).
- Variable types: i.e. are numerical variables coded as string variables?, Are times and dates coded consistently?, etc.
- Check that your data is in the appropriate format (wide vs. long)

Unusual data

- An OUTLIER is a data point whose response y does not follow the general trend of the rest of the data.
- LEVERAGE: extreme value of the predictor x. With multiple predictors, extreme x values may be particularly high or low for one or more predictors, or may be "unusual" combinations of predictor values
- INFLUENTIAL: a data point that unduly. influences the regression analyses' outputs.

Outliers

- Unless an observation is an obvious error, do not try to replace the outlier randomly.
- Outliers can be eliminated from the analysis. In alternative, winsorizing, replacing all values over or under a given percentile (such as the 99th or 1st percentiles) with the value at that percentile.
- When cleaning data, you can decide not to "treat" the outliers (while you may want to when analyzing the data).
- If you drop or winsorize the outliers, always keep note of what you did and how many observations/values have been dropped or winsorized.

Missing

- Check in the codebook if there exist different codes associated with each reason of not answering, standard survey codes don't know, refusal, not eligible, or missing N/A.
- Try to understand the underlying mechanism of missing data as, as ignoring it and taking further action could bias your results:
  - missing completely at random (MCAR) occurs when the probability that a value is missing is independent of both the observed values and the missing values themselves;
  - missing at random (MAR) when the probability that a value is missing depends only on other observed variables in the dataset, but not on the value of the missing variable itself;
  - missing not at random (MNAR) when the probability that a value is missing depends on the value of the missing variable itself.

Missing

- Before taking action try to understand if it is possible to recover data (i.e., go back to responders).
- If you choose to take further action to address missing values, clearly document in the code the method of imputation (e.g. aggregation, multiple imputation,...).

Missing part 2

- Case-wide deletion of observations with missing data. can be an option for a very large dataset with very few missing values, this approach could potentially work really well.
- Replace missing values with the mean/median/mode value of the feature in which they occur. Drawbacks?
- more advanced methods: multiple imputation, inverse probability weighting, likelihood-based methods, last observation carried forward (LOCF) for longitudinal data, and full Bayesian methods. Unbiased if the data is MAR or MCAR (Little, 2021; Collins et al., 2001). In some specific MNAR situations, some of the above methods may still be valid, depending on what you're trying to estimate (White Carlin, 2010).

Multiple-response questions

The multiple-choice question type allows the respondent to choose one or more options from a list of possible answers.

What are the most important aspects when buying wine? a) price b) label c) quality d) country of origin

ID	wine_price	wine_label	wine_quality	wine_country
1	1	1	0	0
2	0	1	1	0
3	0	0	1	1
4	1	1	0	1
5	1	0	0	0

Data is recorded in multiple columns, with one column per answer option. Each variable has two possible values, which indicate whether or not the response was selected by the survey taker.

Data Format: long vs wide

"Long" format

country	year	metric
x	1960	10
x	1970	13
x	2010	15
y	1960	20
y	1970	23
y	2010	25
z	1960	30
z	1970	33
z	2010	35

"Wide" format

country	yr1960	yr1970	yr2010
x	10	13	15
y	20	23	25
z	30	33	35

long vs wide - multinomial logit

- Revealed preferences data: observed choices of individuals for, say, a transport mode (car, plane and train),
- Stated preferences data: individuals face a virtual situation of choice, for example the choice between three train tickets with different characteristics.

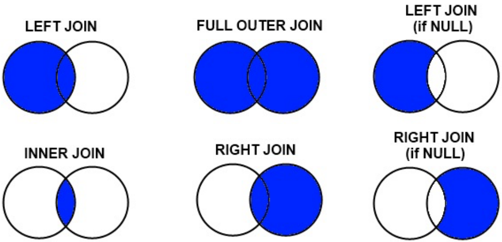
individual	mode	choice	wait	vcost	travel	gcost	income
1	air	no	69	59	100	70	35
1	train	no	34	31	372	71	35
1	bus	no	35	25	417	70	35
1	car	yes	0	10	180	30	35
2	air	no	64	58	68	68	30
2	train	no	44	31	354	84	30

Data merging and appending

- Data merging: consolidates multiple datasets into a single and unified data.
- Data appending: enriches a dataset by adding value-added information into existing records from external sources.

Data merging and appending

Left outer	all left dataset rows, and matching rows from the right dataset. Non-matching rows from the right are excluded.
Inner	only rows that have a match in both datasets.
Right outer	all right dataset rows, and matching rows from the left dataset. Non-matching rows from the left are excluded.
Full outer	all rows from both datasets.



Cross section

- Cross-section,
- Time series,
- Panels.

Cross Section

	id	city	gender	age	sat
1	1	Los Angeles	Female	30	2263
2	2	Sedona	Female	19	2006
3	3	Elmira	Male	26	2221
4	4	Lackawana	Male	33	1716
5	5	Defiance	Male	37	1701
6	6	Tel Aviv	Male	25	1786
7	7	Cimax	Male	39	1577
8	8	Liberal	Female	21	1842
9	9	Montreal	Female	18	1813
10	10	New York	Female	33	2041
11	11	Hot Coffe	Male	18	1787
12	12	Java	Female	38	1513
13	13	Varna	Male	30	1637
14	14	Moscow	Male	30	1512
15	15	Drunkard Creek	Female	21	1338
16	16	Mexican Hat	Female	18	1821
17	17	Amsterdam	Female	19	1494
18	18	Mexico	Female	31	2248
19	19	Caracas	Female	18	2252
20	20	San Juan	Male	33	1923

Time series

Consumption expenditure (X) and Gross domestic product (Y), Both in 1992 billions of dollars

Year	X	X
1982	3081.5	4620.3
1983	3240.6	4803.7
1984	3407.6	5140.1
1985	3566.5	5323.5
1986	3708.7	5487.7
1987	3822.3	5649.5
1988	3972.7	5865.2
1989	4064.6	6062
1990	4132.2	6136.3
1991	4105.8	6079.4
1992	4219.8	6244.4
1993	4343.6	6389.6
1994	4486	6610.7
1995	4595.3	6742.1
1996	4714.1	6928.4

Panel

wave	country	hid	pid	age	sex	healthstat
1990	9	101	1101	46	F	3
1991	1	101	1101	47	F	3
1992	12	101	1101	48	F	2
1993	12	101	1101	49	F	2
1994	9	101	1101	50	F	2
1990	1	201	1102	21	M	5
1991	12	201	1102	22	M	5
1992	9	201	1102	23	M	4
1993	9	201	1102	24	M	5
1994	12	301	1102	25	M	3